TRABAJO FINAL:

ECUACIÓN DE CALOR

-

ESQUEMA IMPLÍCITO

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1. Introducción

En este problema, el objetivo principal es determinar cómo se distribuye el calor a lo largo de una barra de longitud L a medida que transcurre el tiempo t. Para lograrlo, utilizamos las herramientas matemáticas correspondientes a las ecuaciones de conducción de calor, enfocándonos específicamente en la implementación del método implícito. Este enfoque nos permitirá resolver el problema de manera eficiente y estable, derivando finalmente una matriz cuyos vectores han sido calculados a partir de la factorización LU en su forma maximal.

2. Problema matemático

Los esquemas de diferencias finitas para la resolución de ecuaciones en derivadas parciales están basados en la discretización del dominio, de la función, y de sus derivadas parciales. A continuación proporcionamos las fórmulas que utilizaremos a lo largo de los enunciados.

Consideremos una función $u(x,t)$ definida sobre un dominio $\Omega = [0,L] \times [0,T]$. Construimos la malla equidistribuida

$$\{(x_i, t_j) : 0 \leq i \leq I,\ 0 \leq j \leq J\},$$

donde $x_0 = 0$ y $x_i = ih$ para $i = 1, 2, \ldots, I$, y donde $t_0 = 0$, y $t_j = jk$ para $j = 1, 2, \ldots J$, con $h = L/I$ y $k = T/J$. Además, discretizamos la función sobre la malla según $u_{i,j} = u(x_i, t_j)$.

Las derivadas parciales de primer y segundo orden se aproximarán por las siguientes expresiones en diferencias finitas:

$$\begin{aligned} &u_t(x_i,t_j) = \frac{u_{i,j+1} - u_{i,j}}{k}, \quad u_t(x_i,t_j) = \frac{u_{i,j} - u_{i,j-1}}{k}, \quad u_t(x_i,t_j) = \frac{u_{i,j+1} - u_{i,j-1}}{2k}, \\ &u_{tt}(x_i,t_j) = \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{k^2}, \qquad u_{xx}(x_i.t_j) = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2}. \end{aligned} \tag{1}$$

Para las derivadas de primer orden presentamos tres aproximaciones diferentes que denominamos avanzadas, atrasadas o centradas.

Imaginemos una barra de grosor infinitesimal y de longitud L, cuyos extremos están fijados a dos fuentes de calor. En este caso, definimos u(x,t) como la función que describe cómo se distribuye la temperatura a lo largo de la barra en un instante de tiempo t. Esta función está definida en el dominio Ω=[0,L]×[0,T]

Supongamos que conocemos la distribución inicial de la temperatura, a la que llamamos f(x). Además, las fuentes de calor en los extremos de la barra tienen distribuciones de temperatura l(t) y r(t), correspondientes al extremo izquierdo y derecho, respectivamente. Entonces, la evolución de la temperatura en la barra está determinada por la ecuación de calor, junto con las condiciones iniciales y de frontera:

$$
\begin{aligned}
&u_t = u_{xx},\\
&u(x,0) = f(x),\\
&u(0,t) = l(t),\\
&u(L,t) = r(t).
\end{aligned}
$$

El problema nos propone calcular la evolución de la temperatura de la barra para los valores L = 1, T = 1, I = 25, J = 50, con las condiciones iniciales y de frontera f(x) = x(x − 1), l(t) = t, r(t) = t

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Vamos a describir el esquema implícito que utilizaremos. Primero, definimos el parámetro $\alpha = \frac{k}{h^2}$

Además de las constantes mencionadas, necesitamos construir una matriz específica. Esta matriz tendrá la forma siguiente:

- En su diagonal principal, los elementos serán $1+2\cdot\alpha$.
- En la diagonal superior e inferior, los elementos serán $-\alpha$.

De esta manera, la matriz quedará estructurada como:

$$\begin{pmatrix} 1+2\alpha & -\alpha & 0 & \dots & 0 \\ -\alpha & 1+2\alpha & \ddots & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -\alpha \\ 0 & \dots & 0 & -\alpha & 1+2\alpha \end{pmatrix}$$

Ahora lo que haremos es resolver el siguiente sistema a través de la factorización LU maximal para obtener todos los vectores que corresponde a la matriz U que van desde el 2 hasta J+1:

$$\begin{pmatrix} 1+2\alpha & -\alpha & 0 & \dots & 0 \\ -\alpha & 1+2\alpha & \ddots & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -\alpha \\ 0 & \dots & 0 & -\alpha & 1+2\alpha \end{pmatrix} \begin{pmatrix} w_1 \\ \vdots \\ w_i \\ \vdots \\ w_{I-1} \end{pmatrix} = \begin{pmatrix} u_{1,j} + \alpha l(t_{j+1}) \\ \vdots \\ u_{i,j} \\ \vdots \\ u_{I-1,j} + \alpha r(t_{j+1}) \end{pmatrix}$$

Lo que haremos es calcular solo una vez el LUPM de la matriz alpha, y los términos independientes que necesitaremos irán cambiando porque usaremos los vectores u(i,j-1) para usarlos como términos independientes para la resolución y obtención de los vectores u(i,j) .

1. Primero, determinaremos las fronteras. En el primer bucle *for*, calcularemos el valor de $u(x, 0)$, y en el segundo bucle *for* evaluaremos dos condiciones: $u(0, t)$ y $u(L, t)$.

```
for i = 1:I+1
  U(i, 1) = f(x_grid(i));
endfor

for j = 1:J+1
  U(1, j) = l(t_grid(j));
  U(I+1, j) = r(t_grid(j));
endfor
```

```
function [r] = r(t)
  r = t;
endfunction
```

```
function [l] = l(t)
  l = t;
endfunction
```

```
function [f] = f(x)
  f = x*(x-1);
endfunction
```

Después de haber resuelto las fronteras, procederemos a calcular el resto, los vectores que van desde j=2 hasta J+1, y luego realizaremos la descomposición de B el cual es la matriz α.

```
function [L,U,P,Q] = descompLUPM(A)
```

Una vez que hayamos realizado los cálculos previos, solo nos quedará determinar los vectores independientes necesarios para resolver el sistema mencionado anteriormente. Así es como quedaría el bucle de resolución:

```
for k = 2:J+1
  independent_term = U(2:I, k-1);
  independent_term(1) = independent_term(1)+alpha*l(t_grid(k));
  independent_term(I-1) = independent_term(I-1)+alpha*r(t_grid(k));

  y = lower_triangular_solver(L,P*independent_term);
  z = upper_triangular_solver(U1,y);
  x = Q*z;
  U(2:I, k) = x;
endfor
```

La resolución final se llevará a cabo utilizando la factorización LU máxima, siguiendo los pasos mostrados en la imagen. Primero, se resuelve el sistema $L * y = P * independent\ term$ sistema $U_1 * z = y$. Después de obtener y, solo nos quedaría decir que la x, sería $x = Q * z$.

3. Resultados

Conjunto de vectores:

Columns 1 through 23:

0	-0.0384	-0.0736	-0.1056	-0.1344	-0.1600	-0.1824	-0.2016	-0.2176	-0.2304	-0.2400
0.2000	0.1752	0.1522	0.1310	0.1116	0.0942	0.0789	0.0656	0.0545	0.0456	0.0389
0.4000	0.3790	0.3596	0.3418	0.3257	0.3114	0.2988	0.2879	0.2789	0.2716	0.2661
0.6000	0.5802	0.5620	0.5454	0.5304	0.5171	0.5054	0.4954	0.4870	0.4804	0.4753
0.8000	0.7806	0.7628	0.7466	0.7320	0.7190	0.7077	0.6979	0.6898	0.6833	0.6784
1.0000	0.9807	0.9631	0.9470	0.9325	0.9197	0.9084	0.8988	0.8907	0.8843	0.8795
1.2000	1.1808	1.1632	1.1471	1.1327	1.1199	1.1087	1.0991	1.0910	1.0846	1.0798
1.4000	1.3808	1.3632	1.3472	1.3328	1.3200	1.3088	1.2992	1.2911	1.2847	1.2799
1.6000	1.5808	1.5632	1.5472	1.5328	1.5200	1.5088	1.4992	1.4912	1.4848	1.4800
1.8000	1.7808	1.7632	1.7472	1.7328	1.7200	1.7088	1.6992	1.6912	1.6848	1.6800
2.0000	1.9808	1.9632	1.9472	1.9328	1.9200	1.9088	1.8992	1.8912	1.8848	1.8800
2.2000	2.1808	2.1632	2.1472	2.1328	2.1200	2.1088	2.0992	2.0912	2.0848	2.0800
2.4000	2.3808	2.3632	2.3472	2.3328	2.3200	2.3088	2.2992	2.2912	2.2848	2.2800
2.6000	2.5808	2.5632	2.5472	2.5328	2.5200	2.5088	2.4992	2.4912	2.4848	2.4800
2.8000	2.7808	2.7632	2.7472	2.7328	2.7200	2.7088	2.6992	2.6912	2.6848	2.6800
3.0000	2.9808	2.9632	2.9472	2.9328	2.9200	2.9088	2.8992	2.8912	2.8848	2.8800
3.2000	3.1808	3.1632	3.1472	3.1328	3.1200	3.1088	3.0992	3.0912	3.0848	3.0800
3.4000	3.3808	3.3632	3.3472	3.3328	3.3200	3.3088	3.2992	3.2912	3.2848	3.2800
3.6000	3.5808	3.5632	3.5472	3.5328	3.5200	3.5088	3.4992	3.4912	3.4848	3.4800
3.8000	3.7808	3.7632	3.7472	3.7328	3.7200	3.7088	3.6992	3.6912	3.6848	3.6800
4.0000	3.9808	3.9632	3.9472	3.9328	3.9200	3.9088	3.8992	3.8912	3.8848	3.8800
4.2000	4.1808	4.1632	4.1472	4.1328	4.1200	4.1088	4.0992	4.0912	4.0848	4.0800
4.4000	4.3808	4.3632	4.3472	4.3328	4.3200	4.3088	4.2992	4.2912	4.2848	4.2800
4.6000	4.5808	4.5632	4.5472	4.5328	4.5200	4.5088	4.4992	4.4912	4.4848	4.4800
4.8000	4.7808	4.7632	4.7472	4.7328	4.7200	4.7088	4.6992	4.6912	4.6848	4.6800
5.0000	4.9808	4.9632	4.9472	4.9328	4.9200	4.9088	4.8992	4.8912	4.8848	4.8800
5.2000	5.1808	5.1632	5.1472	5.1328	5.1200	5.1088	5.0992	5.0912	5.0848	5.0800
5.4000	5.3808	5.3632	5.3472	5.3328	5.3200	5.3088	5.2992	5.2912	5.2848	5.2800
5.6000	5.5808	5.5632	5.5472	5.5328	5.5200	5.5088	5.4992	5.4912	5.4848	5.4800
5.8000	5.7808	5.7632	5.7472	5.7328	5.7200	5.7088	5.6992	5.6912	5.6848	5.6800
6.0000	5.9808	5.9632	5.9472	5.9328	5.9200	5.9088	5.8992	5.8912	5.8848	5.8800
6.2000	6.1808	6.1632	6.1472	6.1328	6.1200	6.1088	6.0992	6.0912	6.0848	6.0800
6.4000	6.3808	6.3632	6.3472	6.3328	6.3200	6.3088	6.2992	6.2912	6.2848	6.2800
6.6000	6.5808	6.5632	6.5472	6.5328	6.5200	6.5088	6.4992	6.4912	6.4848	6.4800
6.8000	6.7808	6.7632	6.7472	6.7328	6.7200	6.7088	6.6992	6.6912	6.6848	6.6800
7.0000	6.9808	6.9632	6.9472	6.9328	6.9200	6.9088	6.8992	6.8912	6.8848	6.8800
7.2000	7.1808	7.1632	7.1472	7.1328	7.1200	7.1088	7.0992	7.0912	7.0848	7.0800
7.4000	7.3808	7.3632	7.3472	7.3328	7.3200	7.3088	7.2992	7.2912	7.2848	7.2800
7.6000	7.5808	7.5632	7.5472	7.5328	7.5200	7.5088	7.4992	7.4912	7.4848	7.4800
7.8000	7.7808	7.7632	7.7472	7.7328	7.7200	7.7088	7.6992	7.6912	7.6848	7.6800
8.0000	7.9808	7.9632	7.9472	7.9328	7.9200	7.9088	7.8992	7.8912	7.8848	7.8800
8.2000	8.1808	8.1632	8.1472	8.1328	8.1200	8.1088	8.0992	8.0912	8.0848	8.0800
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8.8000	8.7808	8.7632	8.7472	8.7328	8.7200	8.7088	8.6992	8.6912	8.6848	8.6800
9.0000	8.9808	8.9632	8.9472	8.9328	8.9200	8.9088	8.8992	8.8912	8.8848	8.8800
9.2000	9.1808	9.1632	9.1472	9.1328	9.1200	9.1088	9.0992	9.0912	9.0848	9.0800
9.4000	9.3808	9.3632	9.3472	9.3328	9.3200	9.3088	9.2992	9.2912	9.2848	9.2800
9.6000	9.5808	9.5632	9.5472	9.5328	9.5200	9.5088	9.4992	9.4912	9.4848	9.4800
9.8000	9.7808	9.7632	9.7472	9.7328	9.7200	9.7088	9.6992	9.6912	9.6848	9.6800
10.0000	9.9808	9.9632	9.9472	9.9328	9.9200	9.9088	9.8992	9.8912	9.8848	9.8800

-0.2464	-0.2496	-0.2496	-0.2464	-0.2400	-0.2304	-0.2176	-0.2016	-0.1824	-0.1600	-0.1344	-0.1056
0.0344	0.0321	0.0321	0.0344	0.0389	0.0456	0.0545	0.0656	0.0789	0.0942	0.1116	0.1310
0.2625	0.2606	0.2606	0.2625	0.2661	0.2716	0.2789	0.2879	0.2988	0.3114	0.3257	0.3418
0.4720	0.4703	0.4703	0.4720	0.4753	0.4804	0.4870	0.4954	0.5054	0.5171	0.5304	0.5454
0.6752	0.6735	0.6735	0.6752	0.6784	0.6833	0.6898	0.6979	0.7077	0.7190	0.7320	0.7466
0.8763	0.8746	0.8746	0.8763	0.8795	0.8843	0.8907	0.8988	0.9084	0.9197	0.9325	0.9470
1.0766	1.0750	1.0750	1.0766	1.0798	1.0846	1.0910	1.0991	1.1087	1.1199	1.1327	1.1471
1.2767	1.2751	1.2751	1.2767	1.2799	1.2847	1.2911	1.2992	1.3088	1.3200	1.3328	1.3472
1.4768	1.4752	1.4752	1.4768	1.4800	1.4848	1.4912	1.4992	1.5088	1.5200	1.5328	1.5472
1.6768	1.6752	1.6752	1.6768	1.6800	1.6848	1.6912	1.6992	1.7088	1.7200	1.7328	1.7472
1.8768	1.8752	1.8752	1.8768	1.8800	1.8848	1.8912	1.8992	1.9088	1.9200	1.9328	1.9472
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2.8768	2.8752	2.8752	2.8768	2.8800	2.8848	2.8912	2.8992	2.9088	2.9200	2.9328	2.9472
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4.0768	4.0752	4.0752	4.0768	4.0800	4.0848	4.0912	4.0992	4.1088	4.1200	4.1328	4.1472
4.2768	4.2752	4.2752	4.2768	4.2800	4.2848	4.2912	4.2992	4.3088	4.3200	4.3328	4.3472
4.4768	4.4752	4.4752	4.4768	4.4800	4.4848	4.4912	4.4992	4.5088	4.5200	4.5328	4.5472
4.6768	4.6752	4.6752	4.6768	4.6800	4.6848	4.6912	4.6992	4.7088	4.7200	4.7328	4.7472
4.8768	4.8752	4.8752	4.8768	4.8800	4.8848	4.8912	4.8992	4.9088	4.9200	4.9328	4.9472
5.0768	5.0752	5.0752	5.0768	5.0800	5.0848	5.0912	5.0992	5.1088	5.1200	5.1328	5.1472
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7.0768	7.0752	7.0752	7.0768	7.0800	7.0848	7.0912	7.0992	7.1088	7.1200	7.1328	7.1472
7.2768	7.2752	7.2752	7.2768	7.2800	7.2848	7.2912	7.2992	7.3088	7.3200	7.3328	7.3472
7.4768	7.4752	7.4752	7.4768	7.4800	7.4848	7.4912	7.4992	7.5088	7.5200	7.5328	7.5472
7.6768	7.6752	7.6752	7.6768	7.6800	7.6848	7.6912	7.6992	7.7088	7.7200	7.7328	7.7472
7.8768	7.8752	7.8752	7.8768	7.8800	7.8848	7.8912	7.8992	7.9088	7.9200	7.9328	7.9472
8.0768	8.0752	8.0752	8.0768	8.0800	8.0848	8.0912	8.0992	8.1088	8.1200	8.1328	8.1472
8.2768	8.2752	8.2752	8.2768	8.2800	8.2848	8.2912	8.2992	8.3088	8.3200	8.3328	8.3472
8.4768	8.4752	8.4752	8.4768	8.4800	8.4848	8.4912	8.4992	8.5088	8.5200	8.5328	8.5472
8.6768	8.6752	8.6752	8.6768	8.6800	8.6848	8.6912	8.6992	8.7088	8.7200	8.7328	8.7472
8.8768	8.8752	8.8752	8.8768	8.8800	8.8848	8.8912	8.8992	8.9088	8.9200	8.9328	8.9472
9.0768	9.0752	9.0752	9.0768	9.0800	9.0848	9.0912	9.0992	9.1088	9.1200	9.1328	9.1472
9.2768	9.2752	9.2752	9.2768	9.2800	9.2848	9.2912	9.2992	9.3088	9.3200	9.3328	9.3472
9.4768	9.4752	9.4752	9.4768	9.4800	9.4848	9.4912	9.4992	9.5088	9.5200	9.5328	9.5472
9.6768	9.6752	9.6752	9.6768	9.6800	9.6848	9.6912	9.6992	9.7088	9.7200	9.7328	9.7472
9.8768	9.8752	9.8752	9.8768	9.8800	9.8848	9.8912	9.8992	9.9088	9.9200	9.9328	9.9472

Columns 24 through 26:

-0.0736	-0.0384	0
0.1522	0.1752	0.2000
0.3596	0.3790	0.4000
0.5620	0.5802	0.6000
0.7628	0.7806	0.8000
0.9631	0.9807	1.0000
1.1632	1.1808	1.2000
1.3632	1.3808	1.4000
1.5632	1.5808	1.6000
1.7632	1.7808	1.8000
1.9632	1.9808	2.0000
2.1632	2.1808	2.2000
2.3632	2.3808	2.4000
2.5632	2.5808	2.6000
2.7632	2.7808	2.8000
2.9632	2.9808	3.0000
3.1632	3.1808	3.2000
3.3632	3.3808	3.4000
3.5632	3.5808	3.6000
3.7632	3.7808	3.8000
3.9632	3.9808	4.0000
4.1632	4.1808	4.2000
4.3632	4.3808	4.4000
4.5632	4.5808	4.6000
4.7632	4.7808	4.8000
4.9632	4.9808	5.0000
5.1632	5.1808	5.2000
5.3632	5.3808	5.4000
5.5632	5.5808	5.6000
5.7632	5.7808	5.8000
5.9632	5.9808	6.0000
6.1632	6.1808	6.2000
6.3632	6.3808	6.4000
6.5632	6.5808	6.6000
6.7632	6.7808	6.8000
6.9632	6.9808	7.0000
7.1632	7.1808	7.2000
7.3632	7.3808	7.4000
7.5632	7.5808	7.6000
7.7632	7.7808	7.8000
7.9632	7.9808	8.0000
8.1632	8.1808	8.2000
8.3632	8.3808	8.4000
8.5632	8.5808	8.6000
8.7632	8.7808	8.8000
8.9632	8.9808	9.0000
9.1632	9.1808	9.2000
9.3632	9.3808	9.4000
9.5632	9.5808	9.6000
9.7632	9.7808	9.8000
9.9632	9.9808	10.0000

Matriz con los resultados:

Columns 1 through 21:

0	0.2000	0.4000	0.6000	0.8000	1.0000	1.2000	1.4000	1.6000	1.8000	2.0000
-0.0384	0.1752	0.3790	0.5802	0.7806	0.9807	1.1808	1.3808	1.5808	1.7808	1.9808
-0.0736	0.1522	0.3596	0.5620	0.7628	0.9631	1.1632	1.3632	1.5632	1.7632	1.9632
-0.1056	0.1310	0.3418	0.5454	0.7466	0.9470	1.1471	1.3472	1.5472	1.7472	1.9472
-0.1344	0.1116	0.3257	0.5304	0.7320	0.9325	1.1327	1.3328	1.5328	1.7328	1.9328
-0.1600	0.0942	0.3114	0.5171	0.7190	0.9197	1.1199	1.3200	1.5200	1.7200	1.9200
-0.1824	0.0789	0.2988	0.5054	0.7077	0.9084	1.1087	1.3088	1.5088	1.7088	1.9088
-0.2016	0.0656	0.2879	0.4954	0.6979	0.8988	1.0991	1.2992	1.4992	1.6992	1.8992
-0.2176	0.0545	0.2789	0.4870	0.6898	0.8907	1.0910	1.2911	1.4912	1.6912	1.8912
-0.2304	0.0456	0.2716	0.4804	0.6833	0.8843	1.0846	1.2847	1.4848	1.6848	1.8848
-0.2400	0.0389	0.2661	0.4753	0.6784	0.8795	1.0798	1.2799	1.4800	1.6800	1.8800
-0.2464	0.0344	0.2625	0.4720	0.6752	0.8763	1.0766	1.2767	1.4768	1.6768	1.8768
-0.2496	0.0321	0.2606	0.4703	0.6735	0.8746	1.0750	1.2751	1.4752	1.6752	1.8752
-0.2496	0.0321	0.2606	0.4703	0.6735	0.8746	1.0750	1.2751	1.4752	1.6752	1.8752
-0.2464	0.0344	0.2625	0.4720	0.6752	0.8763	1.0766	1.2767	1.4768	1.6768	1.8768
-0.2400	0.0389	0.2661	0.4753	0.6784	0.8795	1.0798	1.2799	1.4800	1.6800	1.8800
-0.2304	0.0456	0.2716	0.4804	0.6833	0.8843	1.0846	1.2847	1.4848	1.6848	1.8848
-0.2176	0.0545	0.2789	0.4870	0.6898	0.8907	1.0910	1.2911	1.4912	1.6912	1.8912
-0.2016	0.0656	0.2879	0.4954	0.6979	0.8988	1.0991	1.2992	1.4992	1.6992	1.8992
-0.1824	0.0789	0.2988	0.5054	0.7077	0.9084	1.1087	1.3088	1.5088	1.7088	1.9088
-0.1600	0.0942	0.3114	0.5171	0.7190	0.9197	1.1199	1.3200	1.5200	1.7200	1.9200
-0.1344	0.1116	0.3257	0.5304	0.7320	0.9325	1.1327	1.3328	1.5328	1.7328	1.9328
-0.1056	0.1310	0.3418	0.5454	0.7466	0.9470	1.1471	1.3472	1.5472	1.7472	1.9472
-0.0736	0.1522	0.3596	0.5620	0.7628	0.9631	1.1632	1.3632	1.5632	1.7632	1.9632
-0.0384	0.1752	0.3790	0.5802	0.7806	0.9807	1.1808	1.3808	1.5808	1.7808	1.9808
0	0.2000	0.4000	0.6000	0.8000	1.0000	1.2000	1.4000	1.6000	1.8000	2.0000

2.2000	2.4000	2.6000	2.8000	3.0000	3.2000	3.4000	3.6000	3.8000	4.0000
2.1808	2.3808	2.5808	2.7808	2.9808	3.1808	3.3808	3.5808	3.7808	3.9808
2.1632	2.3632	2.5632	2.7632	2.9632	3.1632	3.3632	3.5632	3.7632	3.9632
2.1472	2.3472	2.5472	2.7472	2.9472	3.1472	3.3472	3.5472	3.7472	3.9472
2.1328	2.3328	2.5328	2.7328	2.9328	3.1328	3.3328	3.5328	3.7328	3.9328
2.1200	2.3200	2.5200	2.7200	2.9200	3.1200	3.3200	3.5200	3.7200	3.9200
2.1088	2.3088	2.5088	2.7088	2.9088	3.1088	3.3088	3.5088	3.7088	3.9088
2.0992	2.2992	2.4992	2.6992	2.8992	3.0992	3.2992	3.4992	3.6992	3.8992
2.0912	2.2912	2.4912	2.6912	2.8912	3.0912	3.2912	3.4912	3.6912	3.8912
2.0848	2.2848	2.4848	2.6848	2.8848	3.0848	3.2848	3.4848	3.6848	3.8848
2.0800	2.2800	2.4800	2.6800	2.8800	3.0800	3.2800	3.4800	3.6800	3.8800
2.0768	2.2768	2.4768	2.6768	2.8768	3.0768	3.2768	3.4768	3.6768	3.8768
2.0752	2.2752	2.4752	2.6752	2.8752	3.0752	3.2752	3.4752	3.6752	3.8752
2.0752	2.2752	2.4752	2.6752	2.8752	3.0752	3.2752	3.4752	3.6752	3.8752
2.0768	2.2768	2.4768	2.6768	2.8768	3.0768	3.2768	3.4768	3.6768	3.8768
2.0800	2.2800	2.4800	2.6800	2.8800	3.0800	3.2800	3.4800	3.6800	3.8800
2.0848	2.2848	2.4848	2.6848	2.8848	3.0848	3.2848	3.4848	3.6848	3.8848
2.0912	2.2912	2.4912	2.6912	2.8912	3.0912	3.2912	3.4912	3.6912	3.8912
2.0992	2.2992	2.4992	2.6992	2.8992	3.0992	3.2992	3.4992	3.6992	3.8992
2.1088	2.3088	2.5088	2.7088	2.9088	3.1088	3.3088	3.5088	3.7088	3.9088
2.1200	2.3200	2.5200	2.7200	2.9200	3.1200	3.3200	3.5200	3.7200	3.9200
2.1328	2.3328	2.5328	2.7328	2.9328	3.1328	3.3328	3.5328	3.7328	3.9328
2.1472	2.3472	2.5472	2.7472	2.9472	3.1472	3.3472	3.5472	3.7472	3.9472
2.1632	2.3632	2.5632	2.7632	2.9632	3.1632	3.3632	3.5632	3.7632	3.9632
2.1808	2.3808	2.5808	2.7808	2.9808	3.1808	3.3808	3.5808	3.7808	3.9808
2.2000	2.4000	2.6000	2.8000	3.0000	3.2000	3.4000	3.6000	3.8000	4.0000

6.4000	6.6000	6.8000	7.0000	7.2000	7.4000	7.6000	7.8000	8.0000	8.2000
6.3808	6.5808	6.7808	6.9808	7.1808	7.3808	7.5808	7.7808	7.9808	8.1808
6.3632	6.5632	6.7632	6.9632	7.1632	7.3632	7.5632	7.7632	7.9632	8.1632
6.3472	6.5472	6.7472	6.9472	7.1472	7.3472	7.5472	7.7472	7.9472	8.1472
6.3328	6.5328	6.7328	6.9328	7.1328	7.3328	7.5328	7.7328	7.9328	8.1328
6.3200	6.5200	6.7200	6.9200	7.1200	7.3200	7.5200	7.7200	7.9200	8.1200
6.3088	6.5088	6.7088	6.9088	7.1088	7.3088	7.5088	7.7088	7.9088	8.1088
6.2992	6.4992	6.6992	6.8992	7.0992	7.2992	7.4992	7.6992	7.8992	8.0992
6.2912	6.4912	6.6912	6.8912	7.0912	7.2912	7.4912	7.6912	7.8912	8.0912
6.2848	6.4848	6.6848	6.8848	7.0848	7.2848	7.4848	7.6848	7.8848	8.0848
6.2800	6.4800	6.6800	6.8800	7.0800	7.2800	7.4800	7.6800	7.8800	8.0800
6.2768	6.4768	6.6768	6.8768	7.0768	7.2768	7.4768	7.6768	7.8768	8.0768
6.2752	6.4752	6.6752	6.8752	7.0752	7.2752	7.4752	7.6752	7.8752	8.0752
6.2752	6.4752	6.6752	6.8752	7.0752	7.2752	7.4752	7.6752	7.8752	8.0752
6.2768	6.4768	6.6768	6.8768	7.0768	7.2768	7.4768	7.6768	7.8768	8.0768
6.2800	6.4800	6.6800	6.8800	7.0800	7.2800	7.4800	7.6800	7.8800	8.0800
6.2848	6.4848	6.6848	6.8848	7.0848	7.2848	7.4848	7.6848	7.8848	8.0848
6.2912	6.4912	6.6912	6.8912	7.0912	7.2912	7.4912	7.6912	7.8912	8.0912
6.2992	6.4992	6.6992	6.8992	7.0992	7.2992	7.4992	7.6992	7.8992	8.0992
6.3088	6.5088	6.7088	6.9088	7.1088	7.3088	7.5088	7.7088	7.9088	8.1088
6.3200	6.5200	6.7200	6.9200	7.1200	7.3200	7.5200	7.7200	7.9200	8.1200
6.3328	6.5328	6.7328	6.9328	7.1328	7.3328	7.5328	7.7328	7.9328	8.1328
6.3472	6.5472	6.7472	6.9472	7.1472	7.3472	7.5472	7.7472	7.9472	8.1472
6.3632	6.5632	6.7632	6.9632	7.1632	7.3632	7.5632	7.7632	7.9632	8.1632
6.3808	6.5808	6.7808	6.9808	7.1808	7.3808	7.5808	7.7808	7.9808	8.1808
6.4000	6.6000	6.8000	7.0000	7.2000	7.4000	7.6000	7.8000	8.0000	8.2000

Columns 43 through 51:

8.4000	8.6000	8.8000	9.0000	9.2000	9.4000	9.6000	9.8000	10.0000
8.3808	8.5808	8.7808	8.9808	9.1808	9.3808	9.5808	9.7808	9.9808
8.3632	8.5632	8.7632	8.9632	9.1632	9.3632	9.5632	9.7632	9.9632
8.3472	8.5472	8.7472	8.9472	9.1472	9.3472	9.5472	9.7472	9.9472
8.3328	8.5328	8.7328	8.9328	9.1328	9.3328	9.5328	9.7328	9.9328
8.3200	8.5200	8.7200	8.9200	9.1200	9.3200	9.5200	9.7200	9.9200
8.3088	8.5088	8.7088	8.9088	9.1088	9.3088	9.5088	9.7088	9.9088
8.2992	8.4992	8.6992	8.8992	9.0992	9.2992	9.4992	9.6992	9.8992
8.2912	8.4912	8.6912	8.8912	9.0912	9.2912	9.4912	9.6912	9.8912
8.2848	8.4848	8.6848	8.8848	9.0848	9.2848	9.4848	9.6848	9.8848
8.2800	8.4800	8.6800	8.8800	9.0800	9.2800	9.4800	9.6800	9.8800
8.2768	8.4768	8.6768	8.8768	9.0768	9.2768	9.4768	9.6768	9.8768
8.2752	8.4752	8.6752	8.8752	9.0752	9.2752	9.4752	9.6752	9.8752
8.2752	8.4752	8.6752	8.8752	9.0752	9.2752	9.4752	9.6752	9.8752
8.2768	8.4768	8.6768	8.8768	9.0768	9.2768	9.4768	9.6768	9.8768
8.2800	8.4800	8.6800	8.8800	9.0800	9.2800	9.4800	9.6800	9.8800
8.2848	8.4848	8.6848	8.8848	9.0848	9.2848	9.4848	9.6848	9.8848
8.2912	8.4912	8.6912	8.8912	9.0912	9.2912	9.4912	9.6912	9.8912
8.2992	8.4992	8.6992	8.8992	9.0992	9.2992	9.4992	9.6992	9.8992
8.3088	8.5088	8.7088	8.9088	9.1088	9.3088	9.5088	9.7088	9.9088
8.3200	8.5200	8.7200	8.9200	9.1200	9.3200	9.5200	9.7200	9.9200
8.3328	8.5328	8.7328	8.9328	9.1328	9.3328	9.5328	9.7328	9.9328
8.3472	8.5472	8.7472	8.9472	9.1472	9.3472	9.5472	9.7472	9.9472
8.3632	8.5632	8.7632	8.9632	9.1632	9.3632	9.5632	9.7632	9.9632
8.3808	8.5808	8.7808	8.9808	9.1808	9.3808	9.5808	9.7808	9.9808
8.4000	8.6000	8.8000	9.0000	9.2000	9.4000	9.6000	9.8000	10.0000

Plot de los resultados:

